

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Module 8

Random Access Channel (RACH)

کانال دسترسی تصادفی (RACH)

PRACH PREAMBLES

COLLISION



Lecture Notes

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8.1 Why Random Access is Needed

When a UE (User Equipment) powers on or transitions from idle mode, it faces fundamental challenges:

Challenge	Description
No uplink timing alignment	UE doesn't know its propagation delay to the base station
No allocated uplink resources	UE has no scheduled grants for transmission
No unique identity	Network hasn't assigned a C-RNTI yet

Scenarios Requiring Random Access

1. **Initial Access** — UE powering on, first connection to network
2. **Handover** — Moving to a new cell, need synchronization with target eNodeB (eNB)
3. **DL Data Arrival (Idle Mode)** — UE paged for incoming data, must re-establish UL sync
4. **UL Data Arrival** — UE in connected mode lost UL synchronization
5. **Positioning** — Timing measurements for location services

Key Insight: Random access is the **ONLY** mechanism for a UE to initiate uplink communication when it has no prior scheduling from the network.

8.2 PRACH (Physical Random Access Channel)

Time-Frequency Location in Resource Grid

The PRACH occupies a specific position in the LTE resource grid:

- **Frequency domain:** 6 consecutive PRBs (1.08 MHz bandwidth)
- **Time domain:** Configurable subframes based on PRACH configuration index
- **Location:** Specified by `prach-FreqOffset` (PRB offset from lowest UL PRB)

PRACH Configuration Index

The configuration index (0–63) determines:

- **Which subframes** carry PRACH opportunities
- **Periodicity** of PRACH occasions (every 1, 2, 5, or 10 ms)
- **Preamble format** used

Config Index	Preamble Format	Subframe(s)	Density
0	0	1	1 per frame
3	0	1, 4, 7	3 per frame
15	0	0–9	10 per frame
48	4 (short)	1	1 per frame

Preamble Formats

Format	Sequence Length (Ts)	CP Length (Ts)	Total Duration	Max Cell Radius
0	24576 (0.8 ms)	3168 (0.1 ms)	~1 ms	~14.5 km
1	24576 (0.8 ms)	21024 (0.68 ms)	~2 ms	~77.3 km
2	2×24576 (1.6 ms)	6240 (0.2 ms)	~2 ms	~29.5 km
3	2×24576 (1.6 ms)	21024 (0.68 ms)	~3 ms	~100.2 km
4 (short)	4096 (0.13 ms)	448 (0.015 ms)	~0.15 ms	~1.4 km

Design Trade-off: Longer CP → larger cell radius but fewer PRACH opportunities per frame.

8.3 Preamble Sequences

Zadoff-Chu (ZC) Sequences

The PRACH preamble uses Zadoff-Chu sequences defined as:

$$x_u(n) = e^{-j\pi un(n+1)/N_{ZC}}$$

where:

- u = root sequence index ($1 \leq u \leq N_{ZC} - 1$)
- N_{ZC} = sequence length (839 for long, 139 for short)

Why Zadoff-Chu?

Property	Benefit
Constant amplitude (zero PAPR)	Maximizes PA efficiency at cell edge
Ideal periodic autocorrelation	Enables precise timing estimation
Zero cross-correlation between roots	Different root → zero interference
Flat frequency spectrum	Robust in frequency-selective channels

64 Preambles Per Cell

Each cell has exactly **64 available preambles**, derived from:

1. **Root Sequence Index** (`rootSequenceIndex`, broadcast in SIB2)
2. **Cyclic Shifts** applied to root sequences

$$\text{Preamble}(v) = \text{CyclicShift}(x_u, v \times N_{CS})$$

Where:

- N_{CS} = cyclic shift size (`zeroCorrelationZoneConfig`)
- Number of preambles per root = $\lfloor N_{ZC}/N_{CS} \rfloor$
- Multiple roots used if one root doesn't provide 64 preambles

Cell Coverage and Cyclic Shift

The cyclic shift N_{CS} determines the **zero-correlation zone**, which must exceed the maximum round-trip delay:

$$N_{CS} > \frac{2d_{\max}}{cT_{s,\text{seq}}}$$

where $d_{\max}$ is the maximum cell radius, c the speed of light (3×10^8 m/s) and $T_{s,\text{seq}}$ the sequence sample duration.

Example: For $N_{CS} = 13$ (format 0):

- Max distinguishable delay = $13 \times (1/1.25 \text{ MHz}) = 10.4 \mu\text{s}$
- Max cell radius = $10.4 \mu\text{s} \times c/2 \approx 1.56 \text{ km}$
- Preambles per root = $\lfloor 839/13 \rfloor = 64 \rightarrow$ single root sufficient

8.4 Contention-Based Random Access (4-Step CBRA)

Procedure Overview

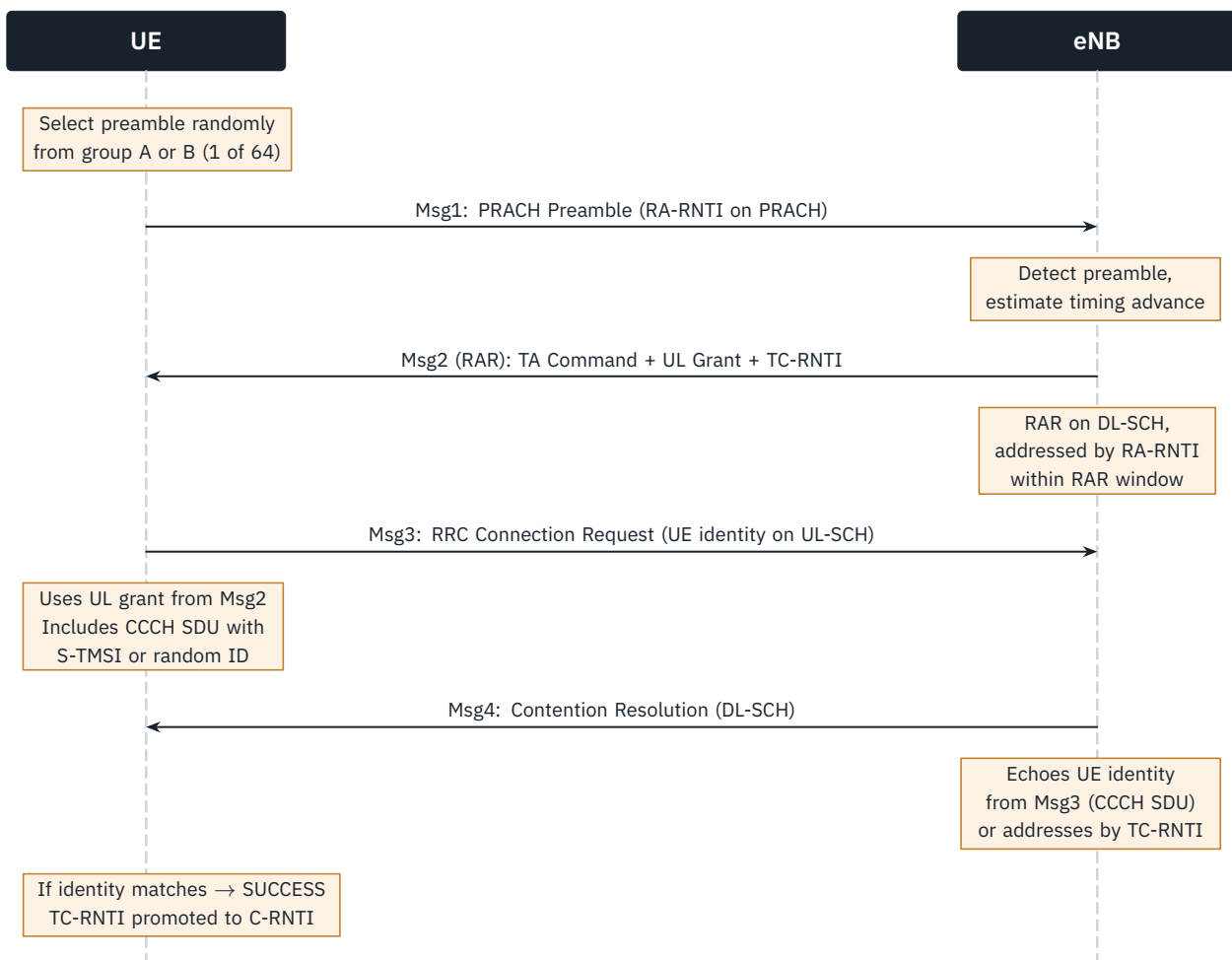


Figure 8.1 Four-step contention-based random access

■ Step-by-Step Details

■ Step 1 – Msg1: Preamble Transmission

- UE selects a **random preamble** from the available set (contention preambles)
- Transmits on PRACH at configured time-frequency resources
- **RA-RNTI** derived from PRACH occasion: $RA-RNTI = 1 + t_id + 10 \times f_id$
 - t_id = subframe index (0–9)
 - f_id = frequency index (0–5)

■ Step 2 – Msg2: Random Access Response (RAR)

RAR contains (per responding preamble):

Field	Bits	Purpose
Timing Advance	11	UL timing correction (0–1282, step = $16 \times T_s$)
UL Grant	20	Resources for Msg3
TC-RNTI	16	Temporary identity
Backoff Indicator	4	Backoff if overloaded

- **RAR Window:** 2–10 subframes (configurable by `ra-ResponseWindowSize`)
- If no RAR received → back off and retry

■ Step 3 – Msg3: Scheduled Transmission

- UE transmits on UL-SCH using the grant from Msg2
- Contains **RRC Connection Request** with:
 - UE identity (S-TMSI or random value)
 - Establishment cause
- HARQ enabled (max 5 retransmissions for Msg3)

■ Step 4 – Msg4: Contention Resolution

- eNB echoes the UE identity received in Msg3
- UE checks if the echoed identity matches its own:
 - **Match** → RACH successful, TC-RNTI becomes C-RNTI
 - **No match** → Collision detected, restart from Step 1 with backoff

8.5 Contention-Free Random Access (CFRA)

When Used

Scenario	Reason
Handover	Fast sync needed, can't afford collision delay
DL data arrival (connected)	eNB knows UE, can pre-assign preamble
Positioning (OTDOA)	Precise timing required
Beam failure recovery	Quick recovery needed

Procedure

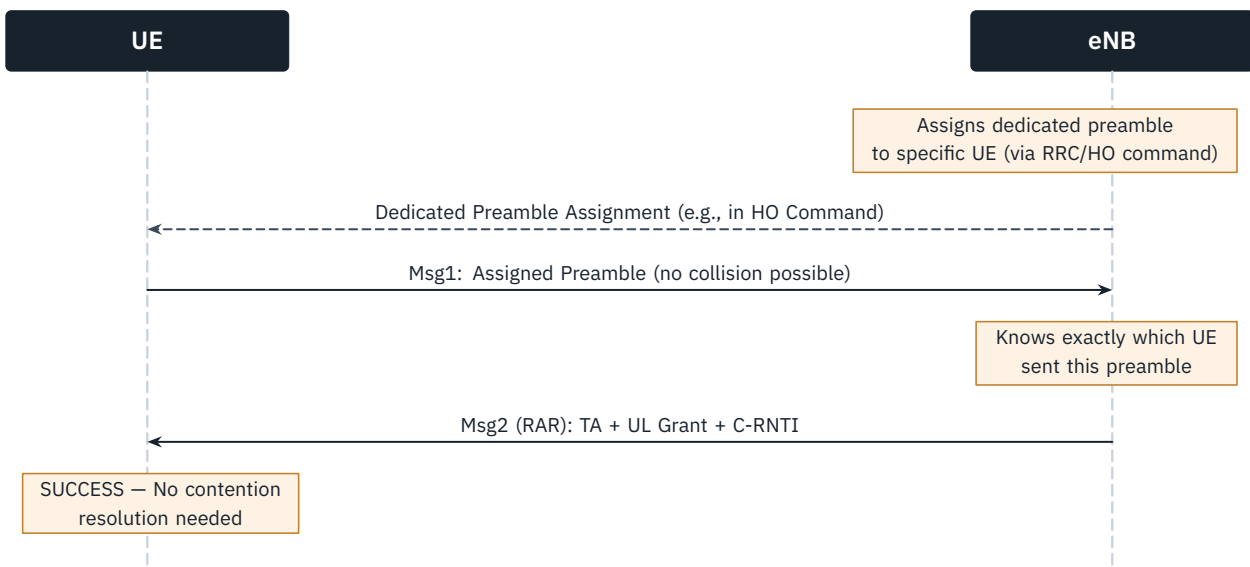


Figure 8.2 Contention-free random access

Key Differences from CBRA

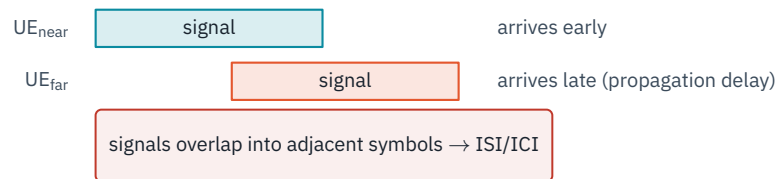
Aspect	Contention-Based	Contention-Free
Steps	4	2
Preamble selection	Random	Assigned by eNB
Collision possible	Yes	No
Latency	~15 ms (success)	~5 ms
Use case	Initial access	Handover, DL data
Preamble pool	Shared (e.g., 54)	Dedicated (e.g., 10)

8.6 Timing Advance (TA)

Why Timing Advance is Needed

In OFDMA uplink (SC-FDMA), all UE signals must arrive at the eNB **aligned within the cyclic prefix** to maintain orthogonality:

Without TA



With TA

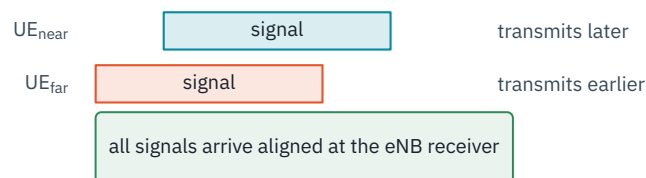


Figure 8.3 Timing advance aligns the uplink signals of near and far UEs

TA Mechanism

Initial TA (from RAR - Msg2):

- 11-bit value: 0–1282
- Step size: $16 \times T_s = 16 \times (1/30.72 \text{ MHz}) \approx 0.52 \mu\text{s}$
- Represents one-way advance applied to UL transmission

TA Maintenance (MAC CE - Timing Advance Command):

- 6-bit value: 0–63
- Applied as: $TA_{\text{new}} = TA_{\text{old}} + (TA_{\text{command}} - 31) \times 16 \times T_s$

- Sent periodically via MAC Control Element
- If no TA update received within `timeAlignmentTimer` → UL sync lost

Cell Radius from Timing Advance

The maximum TA corrects for the **round-trip propagation delay**:

$$\text{Round-trip delay} = \text{TA} \times 16 \times T_s$$

$$\text{One-way delay} = \text{TA} \times 16 \times T_s / 2$$

$$\text{Max distance} = c \times \text{one-way delay}$$

Numerical Example: Maximum Cell Radius

For initial TA (11-bit, max = 1282):

$$\begin{aligned} \text{Max round-trip delay} &= 1282 \times 16 \times T_s = 1282 \times 16 \times \frac{1}{30.72 \times 10^6} \\ &= 1282 \times 0.5208 \mu\text{s} = 667.7 \mu\text{s} \end{aligned}$$

$$\text{Max one-way delay} = 667.7 / 2 = 333.8 \mu\text{s}$$

$$\text{Max cell radius} = 3 \times 10^8 \times 333.8 \times 10^{-6} = 100.1 \text{ km}$$

Verification with preamble format:

Format	Guard Time (CP)	Max RTD	Max Cell Radius
0	99.5 μs ($3168 \times T_s$)	99.5 μs	14.9 km
1	684.4 μs ($21024 \times T_s$)	684.4 μs	102.6 km
2	203.1 μs ($6240 \times T_s$)	203.1 μs	30.5 km
3	684.4 μs ($21024 \times T_s$)	684.4 μs	102.6 km

Note: The actual cell radius is limited by the **smaller** of: TA range OR preamble CP duration.

8.7 Collisions and Backoff

Collision Probability

When N UEs simultaneously attempt random access with 64 available preambles:

Probability that a specific UE collides with at least one other UE:

$$P_{\text{collision}} = 1 - \left(1 - \frac{1}{64}\right)^{N-1}$$

■ Numerical Examples

N (simultaneous UEs)	P(collision) per UE	Expected colliding UEs
2	1.56%	0.03
5	6.10%	0.31
10	13.0%	1.30
20	25.5%	5.10
30	36.8%	11.0
50	54.0%	27.0
64	63.2%	40.4

Example Calculation (N=10):

$$\begin{aligned}
 P(\text{collision}) &= 1 - \left(1 - \frac{1}{64}\right)^{10-1} = 1 - \left(\frac{63}{64}\right)^9 \\
 &= 1 - (0.98438)^9 = 1 - 0.8681 = 0.1319 \approx 13.2\%
 \end{aligned}$$

Collision Scenario with Resolution

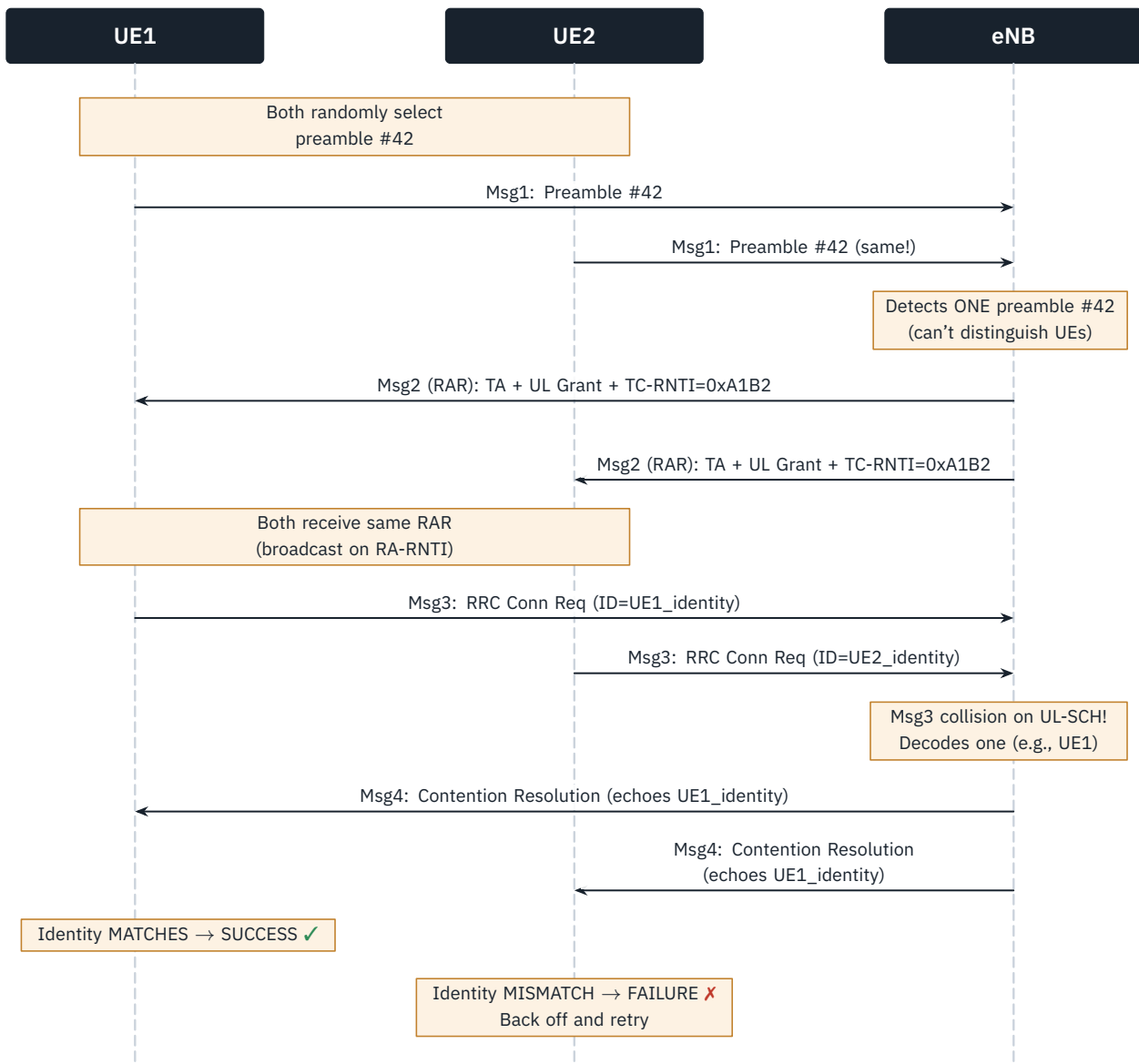


Figure 8.4 Preamble collision and its resolution in Msg4

Backoff Mechanism

When a RACH attempt fails, the UE must wait before retrying:

1. Backoff Indicator (BI): Broadcast in RAR subheader

- Values: 0, 10, 20, 30, 40, 60, 80, 120, 160, 240, 320, 480, 960 ms
- UE selects random backoff: uniform [0, BI]

2. Power Ramping: Each failed attempt increases transmit power

- Step size: `powerRampingStep` (2 dB typical)
- $P_{\text{PRACH}}(\text{attempt}) = P_{\text{initial}} + (\text{attempt} - 1) \times \text{powerRampingStep}$
- Maximum: `preambleInitialReceivedTargetPower` + ramping
- Max attempts: `preambleTransMax` (3, 4, 5, 6, 7, 8, 10, 20, 50, 100, 200)

3. Preamble re-selection: New random preamble chosen for each attempt

RACH Attempt Timeline

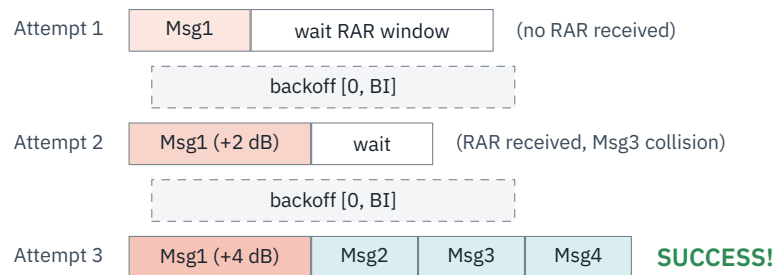


Figure 8.5 RACH attempts with power ramping and backoff

8.8 RACH Overload

Causes of RACH Overload

Cause	Description	Scale
M2M/IoT	Massive simultaneous device activation (e.g., smart meters reporting hourly)	10,000–30,000 devices
Paging storms	Major event triggers mass DL data (emergency alert, popular content)	Thousands of UEs
Disaster/emergency	Everyone calls simultaneously after earthquake/event	Extreme burst
Cell reselection	Many UEs hand over simultaneously (train, stadium)	Hundreds
Power outage recovery	All devices reconnect simultaneously	Thousands

Overload Symptoms

- High collision rate (>50%)
- Excessive preamble retransmissions → increased interference
- RAR capacity exhausted (limited DL resources for RAR)
- Msg3/Msg4 resource starvation
- Access delays from seconds to minutes

■ Solutions

■ Access Class Barring (ACB)

- UEs assigned to Access Classes 0–9 (plus special classes 11–15)
- eNB broadcasts barring factor (0–1) and barring time
- UE draws random number; if > barring factor → barred for barring time
- Spreads access attempts over time

■ Extended Access Barring (EAB)

- For delay-tolerant M2M/IoT devices (AC 0–9)
- eNB broadcasts EAB bitmap (one bit per AC)
- Barred UEs wait for barring to be lifted
- More aggressive than ACB for low-priority traffic

■ Back-off Tuning

- Dynamically increase BI (up to 960 ms) during overload
- Effectively rate-limits RACH attempts
- Trade-off: individual access delay vs. system throughput

■ Additional 3GPP Solutions

Release	Solution	Mechanism
R10	ACB	Probabilistic barring
R11	EAB	IoT/M2M specific barring
R12	ACDC	App-level congestion control
R13	Group paging	Reduce paging-triggered RACH
R14	EDT (Early Data Transmission)	Data in Msg3 → fewer full connections

8.9 5G NR Random Access

Key Enhancements over LTE

2-Step RACH (Rel-16): MsgA + MsgB

Combines the 4-step procedure into 2 steps for **ultra-low latency**:

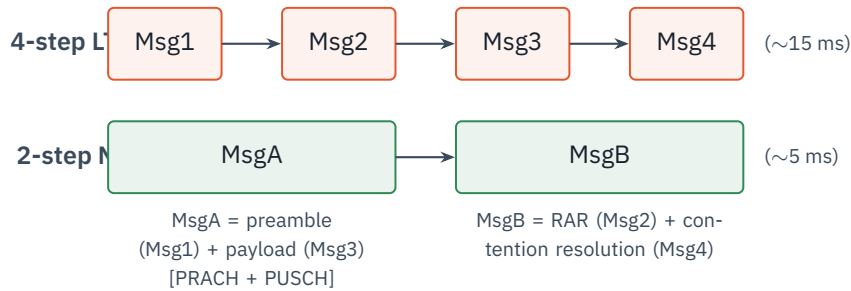


Figure 8.6 Four-step LTE RACH versus two-step NR RACH

When to use 2-step vs 4-step:

- 2-step: Small cells, low load, URLLC requirements
- 4-step: Large cells (need TA before UL data), high load scenarios
- Fallback: If MsgB not received → fall back to 4-step

Beam-Based RACH (SSB-to-RACH Mapping)

In 5G NR with beamforming, the RACH procedure is **beam-aware**:

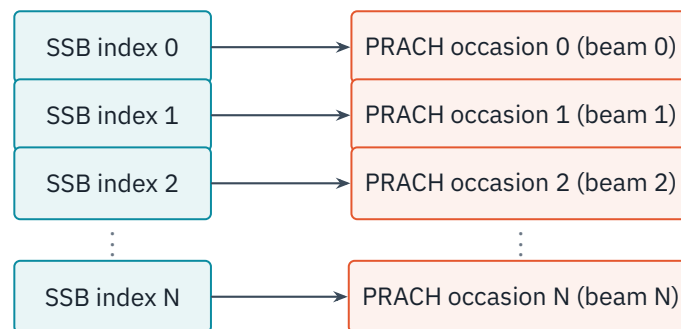


Figure 8.7 SSB-to-RACH mapping: one PRACH occasion per beam

- UE selects **best SSB beam** (highest RSRP)
- Transmits preamble on **associated RACH occasion**
- gNB knows which beam direction the UE is located → **beam correspondence**
- Enables directional RAR transmission

Configuration:

- `ssb-perRACH-OccasionAndCB-PreamblesPerSSB`: maps SSBs to RACH occasions

- Multiple SSBs can share one RACH occasion, or one SSB can span multiple

■ Supplementary Uplink (SUL) RACH

- Lower frequency band (e.g., 700 MHz) for extended coverage
- UE selects SUL PRACH when $rsrp - Threshold_{SSB-SUL}$ condition met
- Separate preamble format and configuration for SUL
- Enables coverage extension without increasing NR UL power

■ NR Preamble Formats

Format	Sequence Length	Subcarrier Spacing	Use Case
0	839 (long)	1.25 kHz	Large cells (FR1)
1	839 (long)	1.25 kHz	Very large cells
2	839 (long)	1.25 kHz	Extreme coverage
3	839 (long)	5 kHz	Medium cells
A1	139 (short)	15/30 kHz	Small cells (FR1)
A2	139 (short)	15/30 kHz	Normal cells (FR1)
A3	139 (short)	15/30 kHz	Normal cells (FR1)
B4	139 (short)	60/120 kHz	FR2 (mmWave)
C0	139 (short)	15/30 kHz	Msg3 early TX
C2	139 (short)	15/30 kHz	Msg3 early TX

■ Beam Failure Recovery (BFR)

A special RACH-like procedure in NR:

- UE detects beam failure (RSRP below threshold for all configured beams)
- UE selects a **new candidate beam** (from measured SSBs/CSI-RS)
- UE transmits **dedicated CFRA preamble** on PRACH occasion of new beam
- gNB responds with BFR response → reconfigure beam pair

8.10 Summary and Key Equations

Critical Formulas

Parameter	Formula
Collision probability	$P_c = 1 - (63/64)^{N-1}$
RA-RNTI	$1 + t_{id} + 10 \times f_{id}$
Timing Advance (initial)	$TA = N_{TA} \times 16 \times T_s$
Max cell radius	$d_{max} = c \times N_{TA,max} \times 16 \times T_s / 2$
Power ramping	$P_n = P_{init} + (n - 1) \times \Delta_{ramp}$
Preambles per root	$\lfloor N_{ZC} / N_{CS} \rfloor$

Design Trade-offs

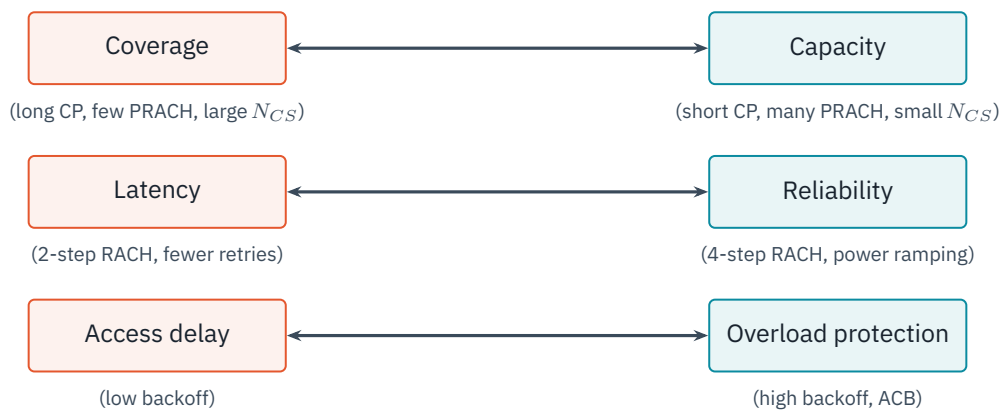


Figure 8.8 RACH design trade-offs

Review Questions

1. Why can't the UE simply transmit on PUSCH without performing RACH first?
2. Calculate the collision probability when 25 UEs attempt simultaneous access.
3. What is the maximum cell radius for PRACH format 3? Show your calculation.
4. Explain why contention-free RACH is used during handover instead of contention-based.
5. A cell has `zeroCorrelationZoneConfig` = 11 ($N_{CS} = 15$). How many root sequences are needed to generate 64 preambles?
6. Describe the beam failure recovery procedure in 5G NR and its relation to RACH.
7. An IoT network has 10,000 devices that simultaneously attempt RACH. Describe two mechanisms to handle this overload.
8. Compare the latency of 2-step NR RACH vs. 4-step LTE RACH. When would you choose each?

References

- 3GPP TS 36.211 §5.7: Physical Random Access Channel (LTE)
- 3GPP TS 36.213 §6: Random Access Procedure (LTE)
- 3GPP TS 36.321 §5.1: Random Access Procedure (MAC)
- 3GPP TS 38.211 §6.3.3: PRACH (NR)
- 3GPP TS 38.213 §8: Random Access Procedure (NR)
- 3GPP TS 38.321 §5.1: Random Access Procedure (NR MAC)